

# MTH 377/577 Convex Optimization

*Instructions: Answer all questions. Max. Marks 25. Time: 1hr.*

1. Find the dual of (i) The cone  $K = \{\theta x | x = \{1, 0, 1\}, \theta \geq 0\}$ ; (ii) The optimization problem  $\max 2x_1 + 3x_2$  s.t.  $x_1 + x_2 \leq 2, 2x_1 - x_2 \leq 3, x_1, x_2 \geq 0$ . (4)

Ans. (i) the dual of  $K$  is the set of vectors  $y$  such that  $y^T x \geq 0$  for any  $x \in K$ . Therefore,  $K^* = \{y \in \mathbb{R}^3 | y_1 + y_3 \geq 0\}$ .

(ii)

$$\begin{aligned} \min \quad & 2y_1 + 3y_2 \\ \text{s.t.} \quad & y_1 + 2y_2 \geq 2 \\ & y_1 - y_2 \geq 3 \\ & y_1, y_2 \geq 0 \end{aligned}$$

2. State whether the following statements are True or False. Provide an explanation/counterexample.

(a) A perspective function of a convex function is always convex. (2)

Ans. True.

**Lemma 2.1.** Let  $f: \mathbb{R}^n \rightarrow [-\infty, \infty]$ . Then, the function  $\check{f}$  defined by

$$\check{f}(x, y) = yf\left(\frac{x}{y}\right), \quad x \in \mathbb{R}^n, y \in \mathbb{R}_+^*$$

is convex if and only if  $f$  is convex.

**Proof.** The *only if* part is obvious (take  $y = 1$ ). Conversely, if  $f$  is convex, then

$$\begin{aligned} & ((1-\lambda)y_1 + \lambda y_2)f\left(\frac{(1-\lambda)x_1 + \lambda x_2}{(1-\lambda)y_1 + \lambda y_2}\right) \\ = & ((1-\lambda)y_1 + \lambda y_2)f\left(\frac{(1-\lambda)y_1}{(1-\lambda)y_1 + \lambda y_2} \frac{x_1}{y_1} + \frac{\lambda y_2}{(1-\lambda)y_1 + \lambda y_2} \frac{x_2}{y_2}\right) \\ \leq & (1-\lambda)y_1 f\left(\frac{x_1}{y_1}\right) + \lambda y_2 f\left(\frac{x_2}{y_2}\right) \end{aligned}$$

for all  $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^n \times \mathbb{R}_+^*$  and all  $\lambda \in (0, 1)$ .

(b) The dual of a proper cone is always convex. (2)

Ans. True. A dual cone is always convex.

- (c) The Lagrangian dual function  $g(\lambda, v)$  is convex. (2)

Ans. False.  $g$  is the pointwise infimum of  $L$  which is affine in  $\lambda, v$  by construction. (show pointwise infimum form of  $L$  as provided in lectures). The pointwise infimum of a family of concave functions is always concave. Since  $L$  is affine, it is both concave and convex. Therefore, the pointwise infimum of  $L$  is concave. Therefore  $g$  is concave.

3. Consider  $A_{n \times n}$ ,  $b_{n \times 1}$  and  $x \in R^n$ . Show that if  $Ax = b$  has a non-negative solution, there is no  $y \in R^n$  such that  $yA \geq 0$  and  $yb < 0$ . Write down the Farkas alternative for the following system of equations and use Farkas Lemma find out if the system has a non-negative solution: (4)

$$\begin{bmatrix} 4 & 1 & 0 \\ 3 & 5 & 3 \\ 0 & 1 & 1 \end{bmatrix} x = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}$$

Ans. Suppose that  $x \in R_+^n$  is a solution for the system  $Ax = b$ . For any  $y \in R^n$ ,  $yAx = yb$ . Suppose for some  $y$  we have  $yA \geq 0$  and  $yb < 0$ . Therefore,  $yAx \neq yb \implies Ax \neq b$  which is a contradiction. Therefore, no such  $y$  can exist.

Farkas' Alternative:  $yA \geq 0$ ,  $yb < 0$  is as follows:

$$4y_1 + 3y_2 \geq 0$$

$$y_1 + 5y_2 + y_3 \geq 0$$

$$3y_2 + y_3 \geq 0$$

$$2y_1 + 3y_2 + y_3 < 0$$

Solving the above for  $y$ , we observe:  $y_1 < 0, y_2 > 0, y_3 > 0$ . From the first inequality,  $3y_2 \geq -4y_1$ . The third inequality implies  $3y_2 + y_3 < -2y_1$ . Since  $y_1 < 0$  and  $y_3 > 0$ , we get a contradiction. Therefore  $y$  does not exist. By Farkas' Lemma, a non-negative solution to the given system exists.

4. Let  $A = \{x \in R^n : \|x\|^2 \leq 4\}$  and  $B = \{(x_1, x_2, \dots, x_n) \in R^n : x_i > 2, \forall i = 1, 2, \dots, n\}$ . Is  $A \cup B$  a convex set? Can you find a separating hyperplane between  $A$  and  $B$ ? If such a hyperplane exists, find it and mention if it is weak or strong? (4)

Ans.  $A \cup B$  is not convex: let  $n = 2$  and consider  $(0, 2) \in A$  and  $(2, 4) \in B$ . Let  $\theta = 0.5$ . Then  $\theta(0, 2) + (1 - \theta)(2, 4) = (1, 3) \notin A \cup B$ . Note that  $A$  and  $B$  are disjoint. Yes, strict separating hyperplane between  $A$  and  $B$  exists.  $\sum_{i=1}^n x_i = 3$  is one such hyperplane.

5. Let  $S_k = \{(x_1, x_2, \dots, x_n) \in R^n : x_i \in [0, k], \forall i = 1, 2, \dots, n\}$  where  $k \in I_+$ . Let  $S = \{S_k | k \in I_+\}$ . Does  $S$  have a minimum? Find a minimal of  $S_2$ . Provide formal arguments for your answer. You may use  $\succeq_{R_+^n}$  where required. (5)

Ans. Yes. We start with  $k = 1$ . Then  $x_i \in [0, 1]$ . A minimal element of  $S_1$  is  $x = (0, 0, \dots, 0)$  since  $y - x \in R_+^n$  for all  $y \in S_1$ . Similarly,  $(0, 0, \dots, 0)$  is also minimal for  $S_2$ . For minimum of  $S$ , we need to pick that  $S_k$  such that  $S \subseteq S_k + R_+^n$  (see if and only if condition for minimum in lecture 8). This holds for all  $k$ . Since  $S_1$  is the smallest such set in  $S$ , we pick it as minimum.

6. Let  $g(x) = \sup\{f_1(x), f_2(x), \dots, f_n(x)\}$  where  $f_1, f_2, \dots, f_n$  are quasiconvex functions with domain  $D \subseteq R^n$ . Is  $g$  quasiconvex? (3)

Ans. Yes.  $f_1(x), f_2(x), \dots, f_n(x)$  are quasiconvex i.e. their sub-level sets are convex. Consider  $\alpha \in R$ . The sub-level sets for  $g$  are:  $L_\alpha = \{x \in D | g(x) \leq \alpha\}$  for each  $\alpha$ . Note that  $L_\alpha = \{x \in D | \sup\{f_1(x), f_2(x), \dots, f_n(x)\} \leq \alpha\} = \{x \in D | f_i(x) \leq \alpha\}$  for  $f_i$  that is supremum. Since  $f_i(x)$  is quasiconvex,  $L_\alpha$  is convex. This holds for each  $\alpha$  and each corresponding supremum  $f$ . Therefore, each  $L_\alpha$  is convex which implies  $g$  is quasiconvex.